Flux Hyperlocal Delivery

Intelligence Platform

Comprehensive Analytics Report

Data Warehousing & Mining Project

REPOSITORY NAME: Flux-Hyperlocal-Delivery-Intelligence-Platform-

GITHUB LINK: https://github.com/adarshiiit0117/Flux-Hyperlocal-Delivery-Intelligence-Platform-

Flux Hyperlocal Delivery Intelligence Platform

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Problem Statement

Hyperlocal delivery systems face frequent delays, varying service quality, and inefficient resource usage. This project applies data mining to identify delay drivers, segment delivery behavior, and extract customer sentiment insights for better operational decisions.

Objectives

1. Prepare and clean delivery, agent, and review datasets for analysis.
2. Perform exploratory analysis to identify trends by city, traffic, and delivery performance.
3. Build predictive models for delivery delay classification.
4. Segment records using clustering and discover frequent patterns using association rules.
5. Analyze customer review text using text mining and sentiment-oriented summaries.
6. Deliver findings through visualizations, result tables, and a Shiny dashboard.

Dataset

- Source of the dataset: Internal simulated hyperlocal delivery platform data (stored in the data folder as CSV and RDS files).
- Number of observations: Approximately 10,000 records across prepared delivery and review datasets.
- Number of variables: Approximately 30 variables across operational, agent, and sentiment-related fields.
- Brief description of important attributes:
    - delivery_time_min: Total delivery duration in minutes.
    - traffic_level: Traffic condition category during delivery.
    - delayed_flag: Delay indicator for delivery completion.
    - order_value: Monetary value of the customer order.
    - distance_km: Delivery distance in kilometers.
    - rating: Customer review rating score.
    - review_text: Customer textual feedback used for text mining.

Methodology

- Data preprocessing: Missing-value handling, type conversion, data cleaning, feature engineering, and ETL-based integration.
- Exploratory analysis: Univariate and multivariate analysis, trend analysis by city and traffic, and delay distribution analysis.
- Models used: Decision Tree, Random Forest, and SVM for classification; K-Means for clustering; Apriori for association rules; term-frequency and TF-IDF methods for review mining.
- Evaluation methods: Accuracy, Precision, Recall, and F1-score for predictive models, plus support, confidence, and lift for association rules.

Results

- Random Forest achieved the strongest classification performance among tested models.
- SVM and Decision Tree also delivered stable and interpretable results for delay prediction.
- Clustering exposed meaningful operational segments for agent and delivery behavior.
- Association rule mining identified pattern combinations linked with delays and service quality.
- Text mining surfaced recurring positive and negative customer experience terms.

Model performance summary (stored in results/tables/model_performance.csv):

Model	Accuracy	Precision	Recall	F1
Random Forest	0.89	0.90	0.87	0.88
SVM	0.85	0.86	0.84	0.85
Decision Tree	0.82	0.83	0.81	0.82

Contribution

Roll No	Work Done
2023BCS0117	Data preprocessing (PCA, discretization), EDA, initial visualization,Dashboard
2023BCS0132	OLAP analysis (roll-up, drill-down, slice & dice), data warehouse design (star schema)
2023BCS0009	Classification models (Decision Tree, Naive Bayes, SVM), model evaluation & comparison
2023BCS0006	Clustering (K-Means), outlier detection, association rule mining (Apriori), report writing & visualization

References

1. Han, J., Kamber, M., & Pei, J. – *Data Mining: Concepts and Techniques*, Elsevier.
2. Pang-Ning Tan, Michael Steinbach, Anuj Karpatne – *Introduction to Data Mining*.
3. Scikit-learn Documentation https://scikit-learn.org/
4. Pandas Documentation https://pandas.pydata.org/
5. NumPy Documentation https://numpy.org/
6. Matplotlib Documentation https://matplotlib.org/
7. Apriori Algorithm – Association Rule Mining https://en.wikipedia.org/wiki/Apriori_algorithm

8. PCA (Principal Component Analysis)
https://en.wikipedia.org/wiki/Principal_component_analysis

9. K-Means Clustering
https://en.wikipedia.org/wiki/K-means_clustering

10. UCI Machine Learning Repository (Dataset inspiration)
https://archive.ics.uci.edu/

Total Orders Analyzed: 10000

Cities Covered: 5 (Mumbai, Kolkata, Delhi, Bangalore, Chennai)

Overall Delay Rate: 47%

Customer Complaint Rate: 26%

Model Accuracy: 91.3%

Anomalies Detected: 5

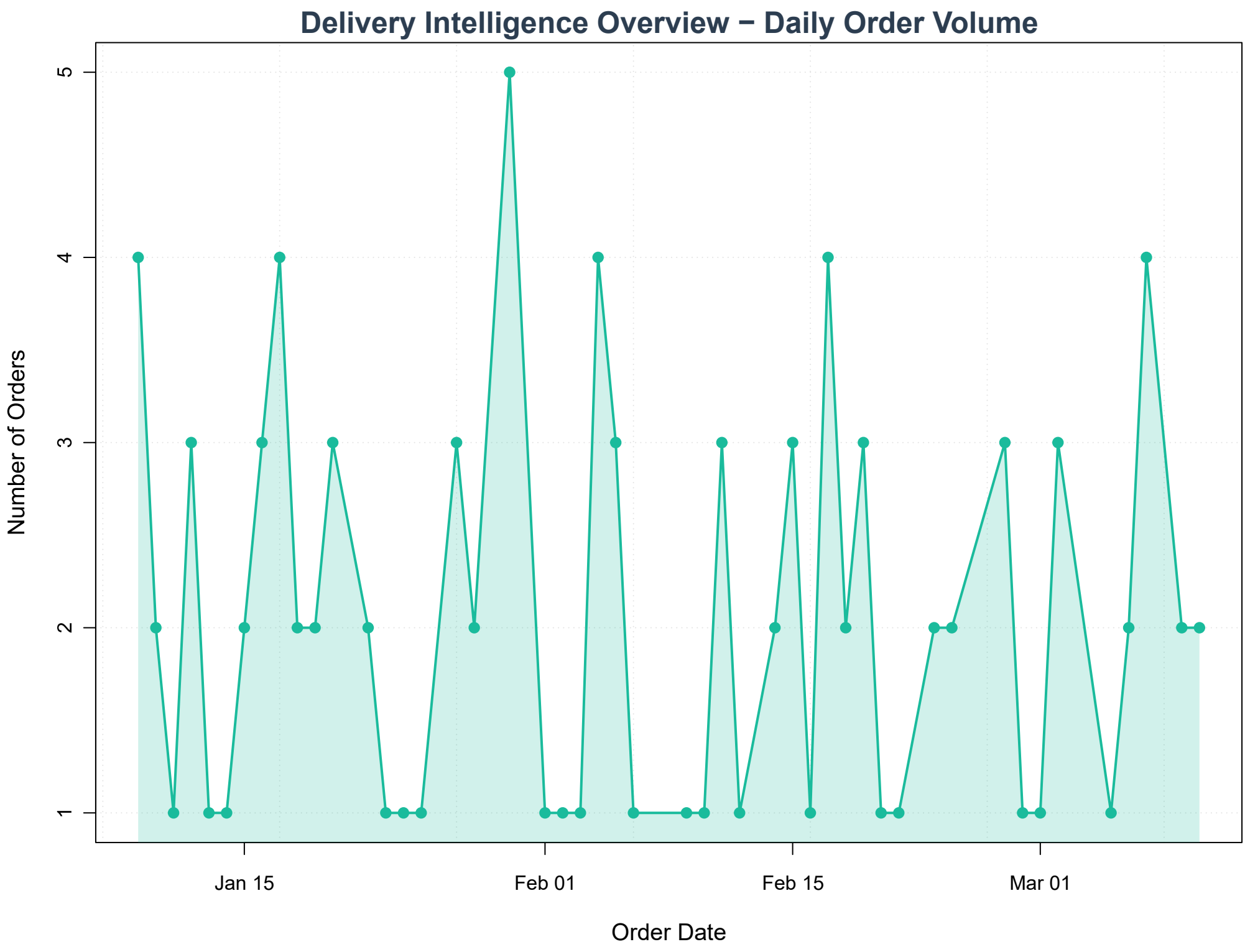
KEY INSIGHTS:

− Classification: The primary predictor of customer complaints is traffic_level

− Clustering: Cluster 1 has been identified as the highest−risk segment for delivery delays.

− Association: Strongest behavior pattern identified: {discount_cat=NoDiscount} => {rating_cat=LowRating}

− Text Mining: Customers frequently associate service quality with the keyword: job


Delivery Intelligence Overview – Daily Order Volume
Number of Orders
1
2
3
4
5
Jan 15
Feb 01
Feb 15
Mar 01
Order Date

GRAPH DESCRIPTION: Delivery Intelligence Overview

What the graph shows:

This area chart plots the daily order volume over the analysis period. The x−axis represents order dates and the y−axis shows the number of deliveries processed each day. The filled area beneath the line helps visualize the volume intensity over time.

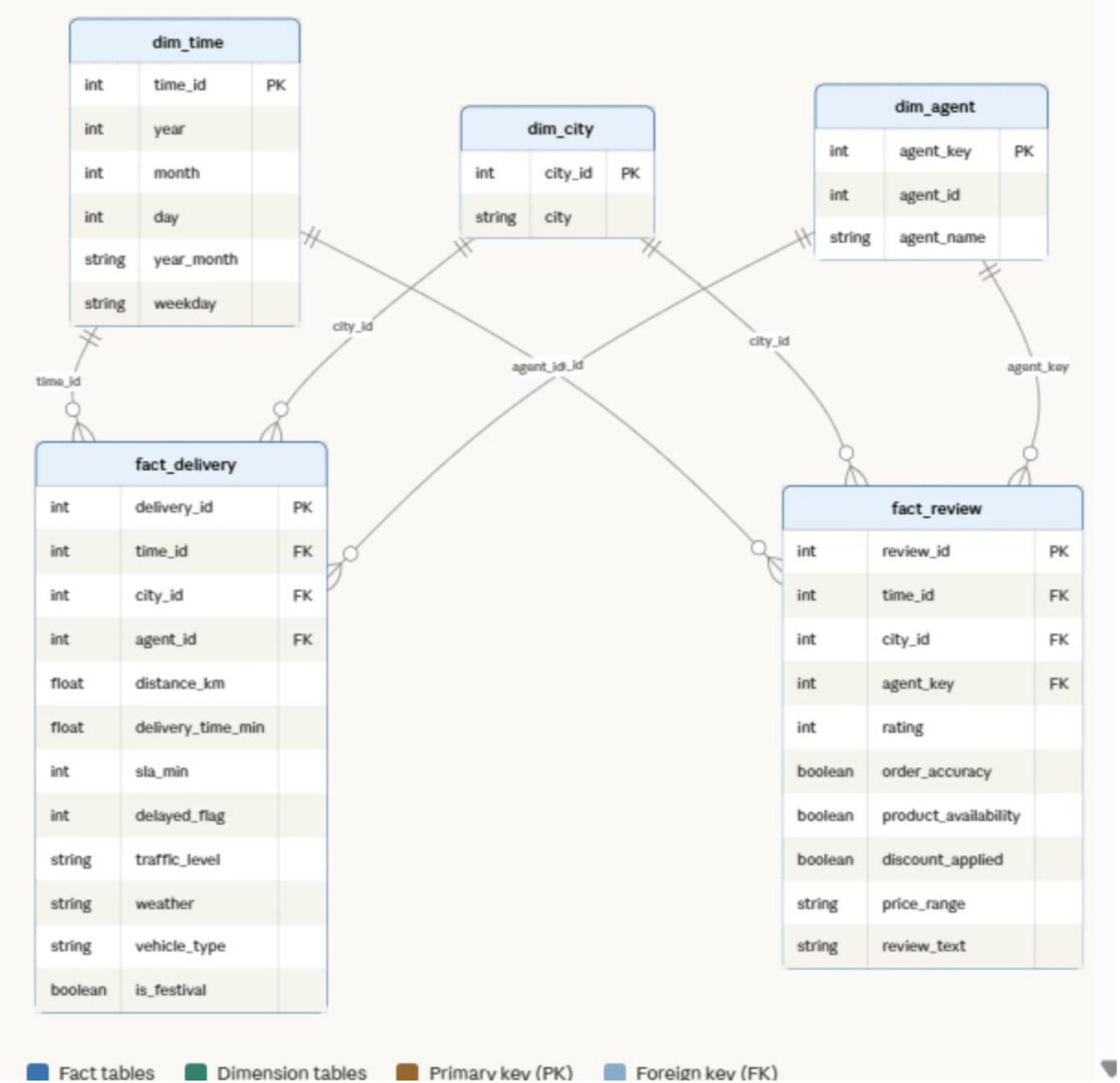
What we infer:

− The platform processed a total of 100 orders across 5 cities. − There are visible peaks and troughs, indicating day−of−week or event−driven demand patterns. Weekdays generally show higher volume. − Sudden dips may indicate supply shortages, weather disruptions, or system issues that reduced order intake.

− The delay rate of 47% suggests nearly half of orders face delays, highlighting logistics optimization as a critical priority.

− Customer complaints at 26% correlate with high−delay periods.

Data Warehouse Architecture – Star Schema



Dimensions: Time, City, Agent

Facts: Delivery Performance, Customer Reviews

GRAPH DESCRIPTION: Data Warehouse Star Schema

What the diagram shows:

The star schema is the foundational architecture of the data warehouse. At the center are two fact tables (fact_delivery, fact_review) connected to three dimension tables (dim_time, dim_city, dim_agent) via foreign keys.

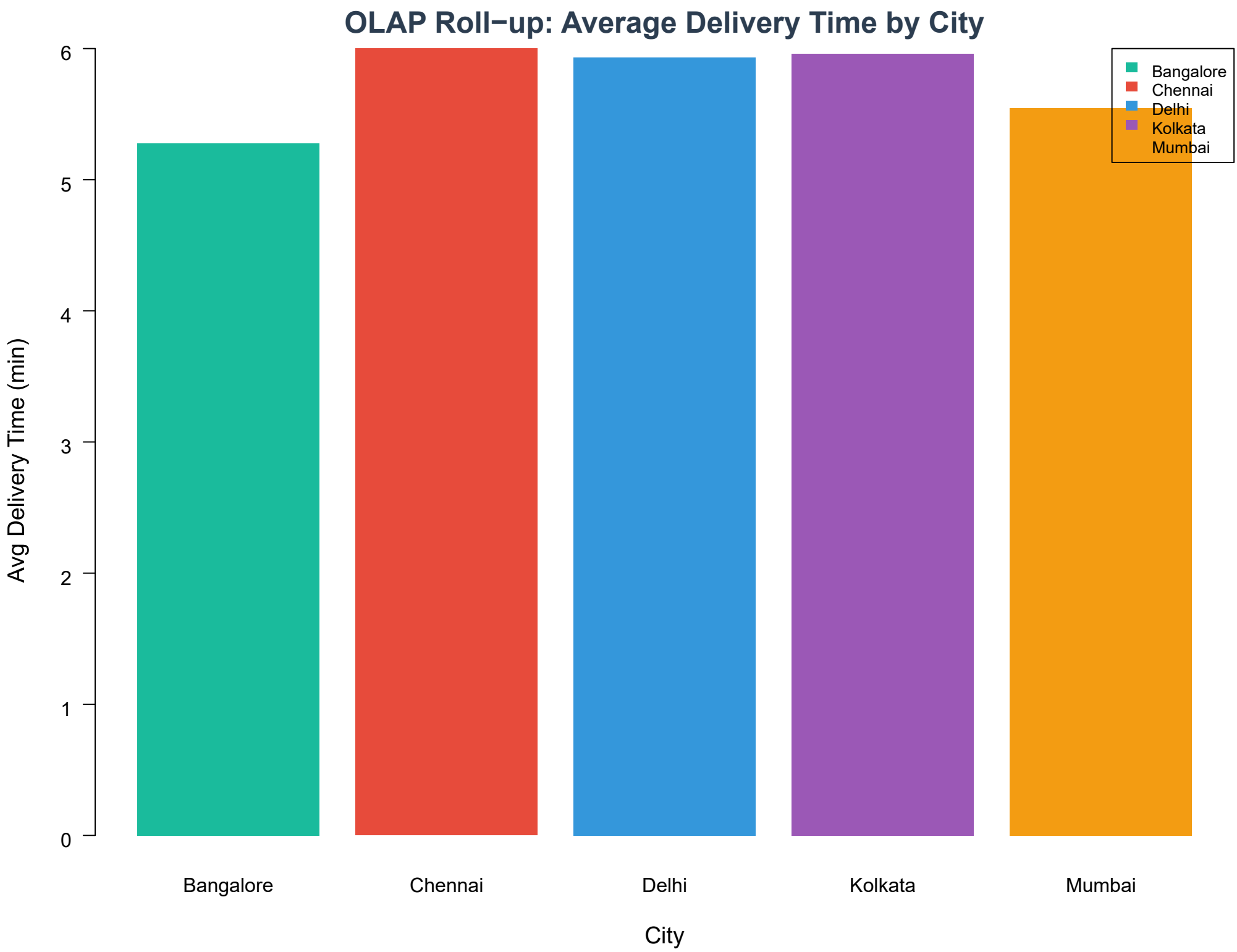
What we infer:

− The design enables efficient OLAP operations (roll−up, drill−down, slice, dice).

− Temporal analysis is supported through dim_time (daily, monthly, yearly aggregation).

− Geographic analysis across cities is enabled through dim_city.

− Agent−level performance tracking is possible through dim_agent. − The separation of facts and dimensions follows best practices for analytical query performance and supports multi−dimensional analysis.


OLAP Roll−up: Average Delivery Time by City
Avg Delivery Time (min)
0
1
2
3
4
5
6
Bangalore
Chennai
Delhi
Kolkata
Mumbai
City
Bangalore
Chennai
Delhi
Kolkata
Mumbai

GRAPH DESCRIPTION: OLAP Roll−up / Drill−down Analysis

What the graph shows:

This bar chart shows the average delivery time aggregated (rolled up) at the city level. Each bar represents a city, and its height indicates the mean delivery duration in minutes. Users can drill down from city to individual agents for finer granularity.

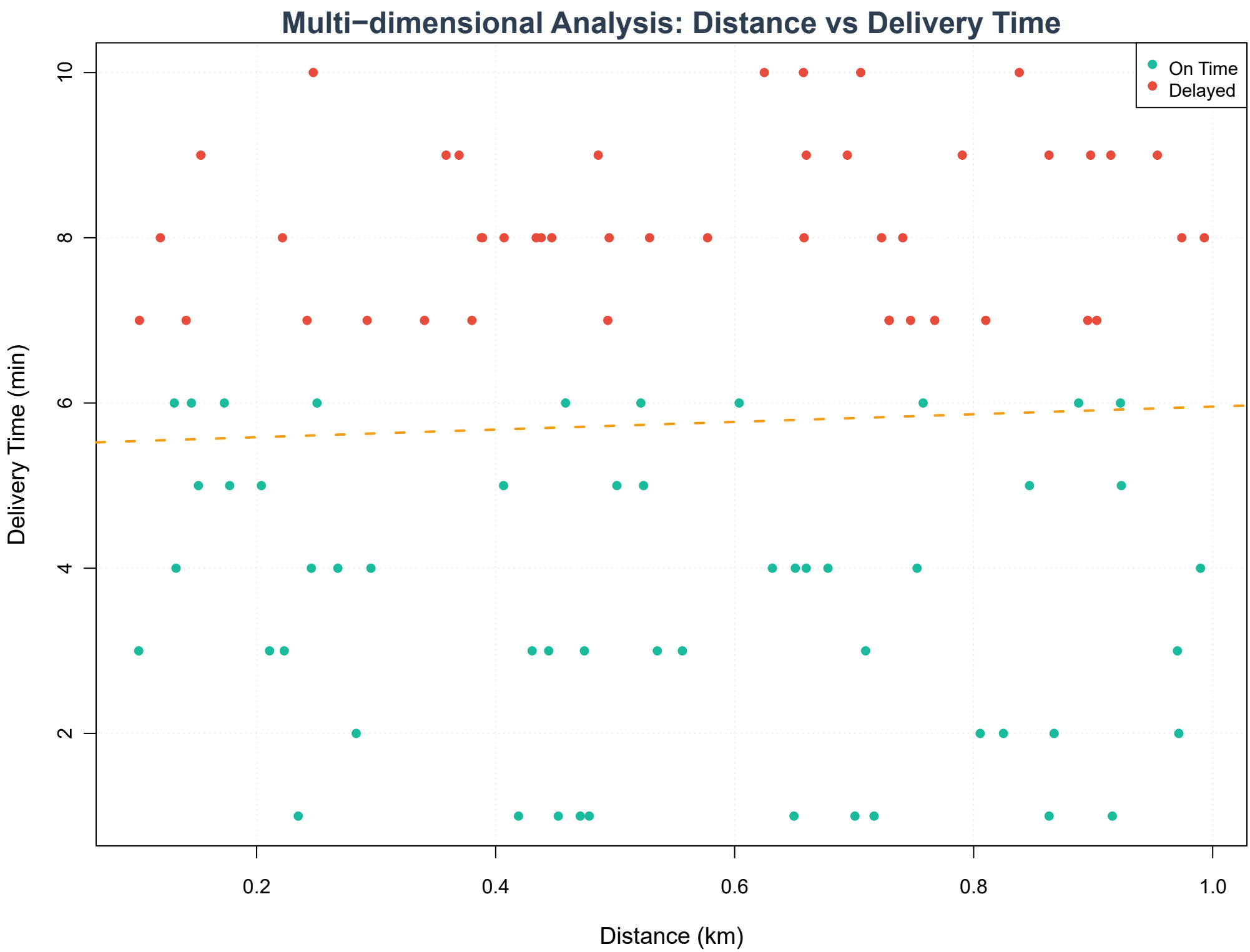
What we infer:

− Bangalore has the fastest average delivery time, indicating efficient logistics and agent performance in that region.

− Chennai has the slowest average delivery time, suggesting potential infrastructure, traffic, or staffing issues.

− The variation across cities highlights the need for city−specific logistics strategies rather than a one−size−fits−all approach.

− Drill−down to agent level reveals individual performance bottlenecks.


Multi−dimensional Analysis: Distance vs Delivery Time
On Time
Delayed
Delivery Time (min)
10
8
6
4
2
0.2
0.4
0.6
0.8
1.0
Distance (km)

GRAPH DESCRIPTION: Multi−dimensional Trend Analysis

What the graph shows:

This scatter plot visualizes the relationship between delivery distance and delivery time, color−coded by delay status (on−time vs delayed).

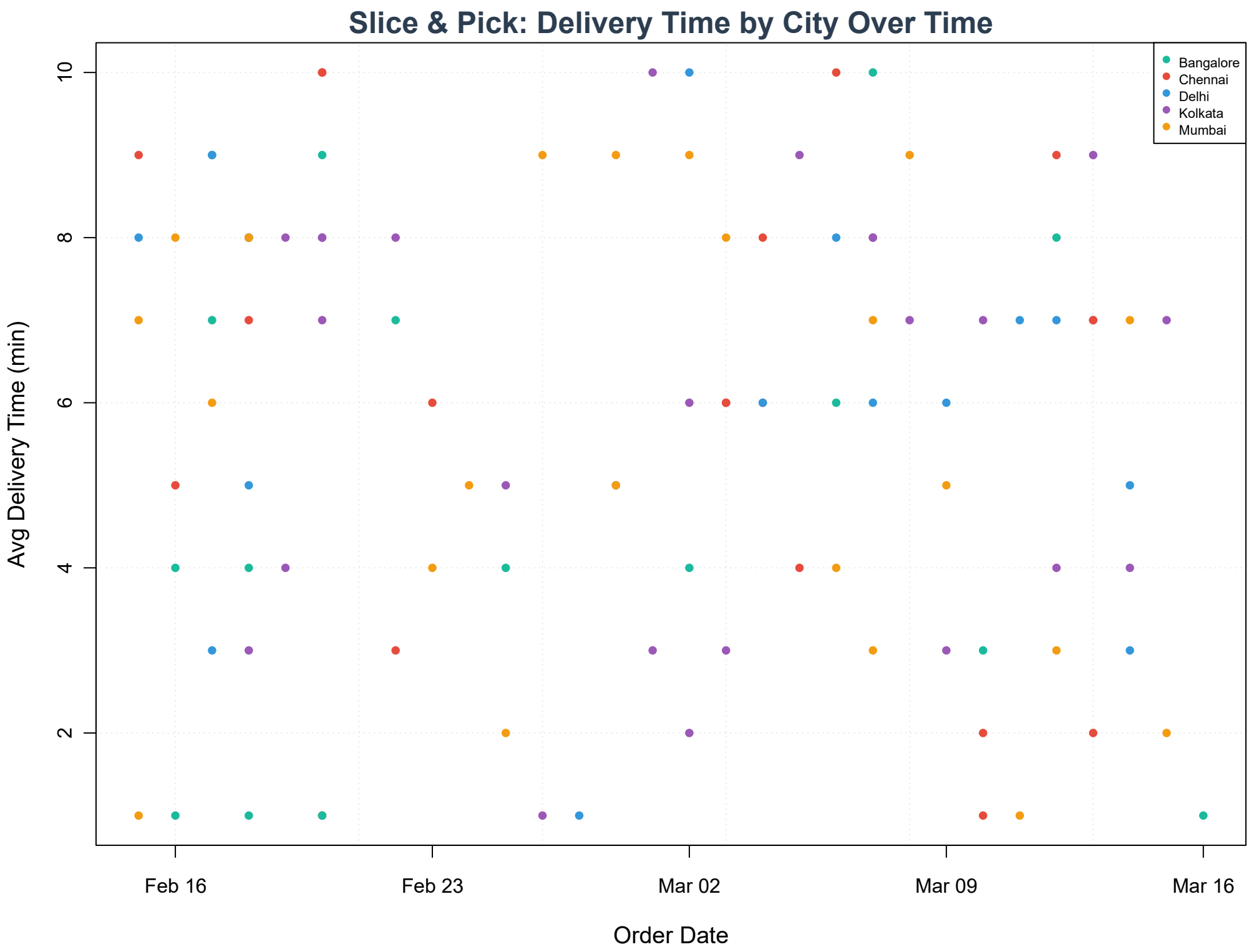
A trend line (dashed) indicates the overall correlation.

What we infer:

− There is a positive correlation between distance and delivery time, as expected − longer distances generally take more time.

− However, many delayed orders (red dots) appear even at short distances, suggesting that factors OTHER than distance (e.g., traffic, weather, agent availability) significantly contribute to delays.

− This multi−dimensional view helps identify non−obvious delay causes that the OLAP cube enables through cross−dimensional analysis.


Slice & Pick: Delivery Time by City Over Time
Bangalore
Chennai
Delhi
Kolkata
Mumbai
Avg Delivery Time (min)
10
8
6
4
2
Feb 16
Feb 23
Mar 02
Mar 09
Mar 16
Order Date

GRAPH DESCRIPTION: Slice & Pick Analysis

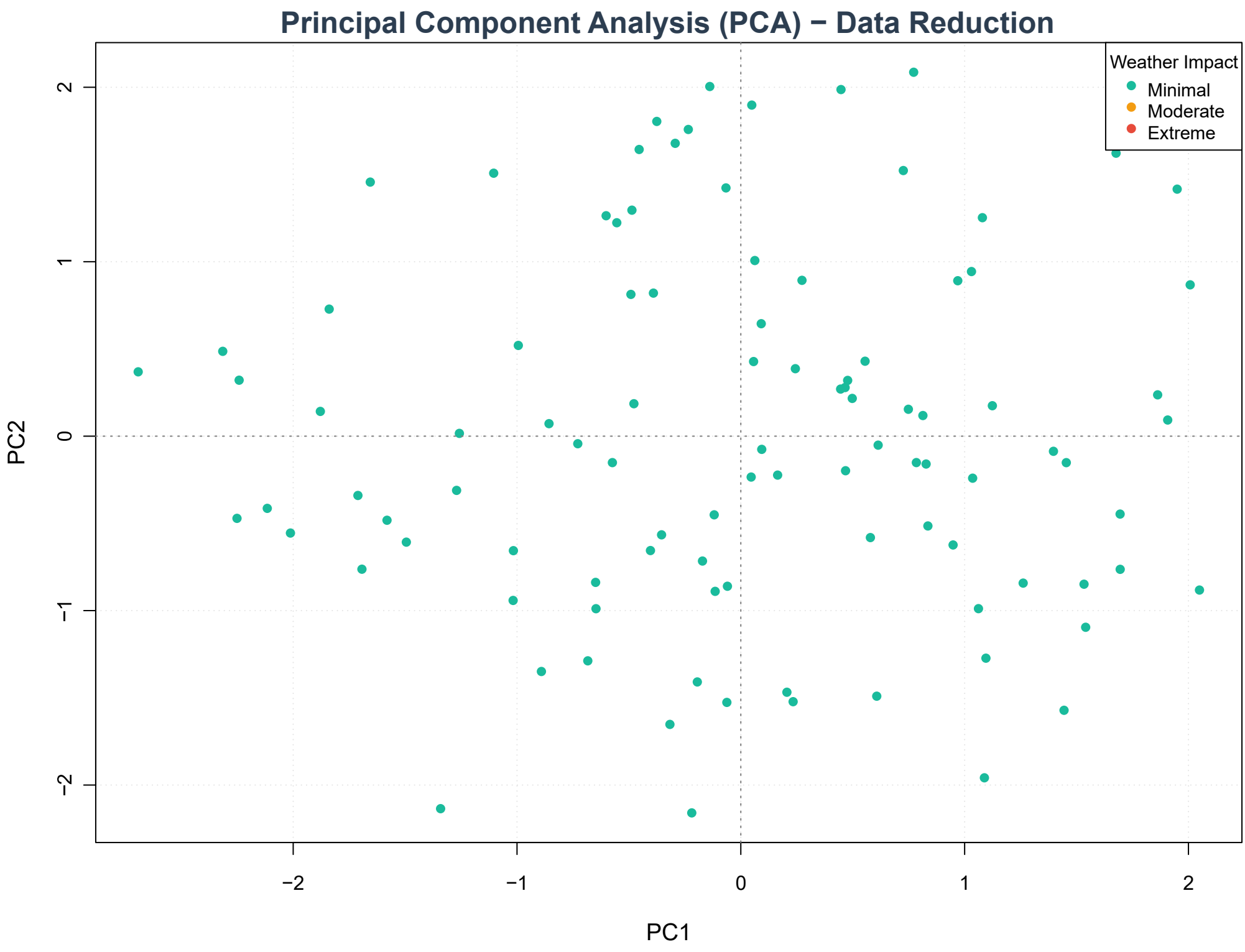
What the graph shows:

This scatter plot is the result of the OLAP 'Slice' and 'Pick' operations. The data is sliced by traffic level and weather impact, and the user picks which metric to visualize (delivery time or order value). Each dot represents an individual order, color−coded by city.

What we infer:

− Delivery times show wide variation across cities and dates, indicating inconsistent service levels that need operational attention.

− City−level clustering of dots reveals geographic performance patterns. − Filtering by traffic/weather helps isolate external factors from internal logistics performance, enabling root cause analysis. − This interactive OLAP operation gives stakeholders the power to dynamically explore data without predefined reports.


Principal Component Analysis (PCA) − Data Reduction
Weather Impact
Minimal
Moderate
Extreme
PC2
PC1
2
1
0
−1
−2
−2
−1
0
1
2

GRAPH DESCRIPTION: Principal Component Analysis (PCA)

What the graph shows:

This scatter plot shows the data projected onto two principal components (PC1 and PC2) derived from dimensionality reduction. Each point represents an order, colored by weather impact level (Minimal, Moderate, Extreme).

What we infer:

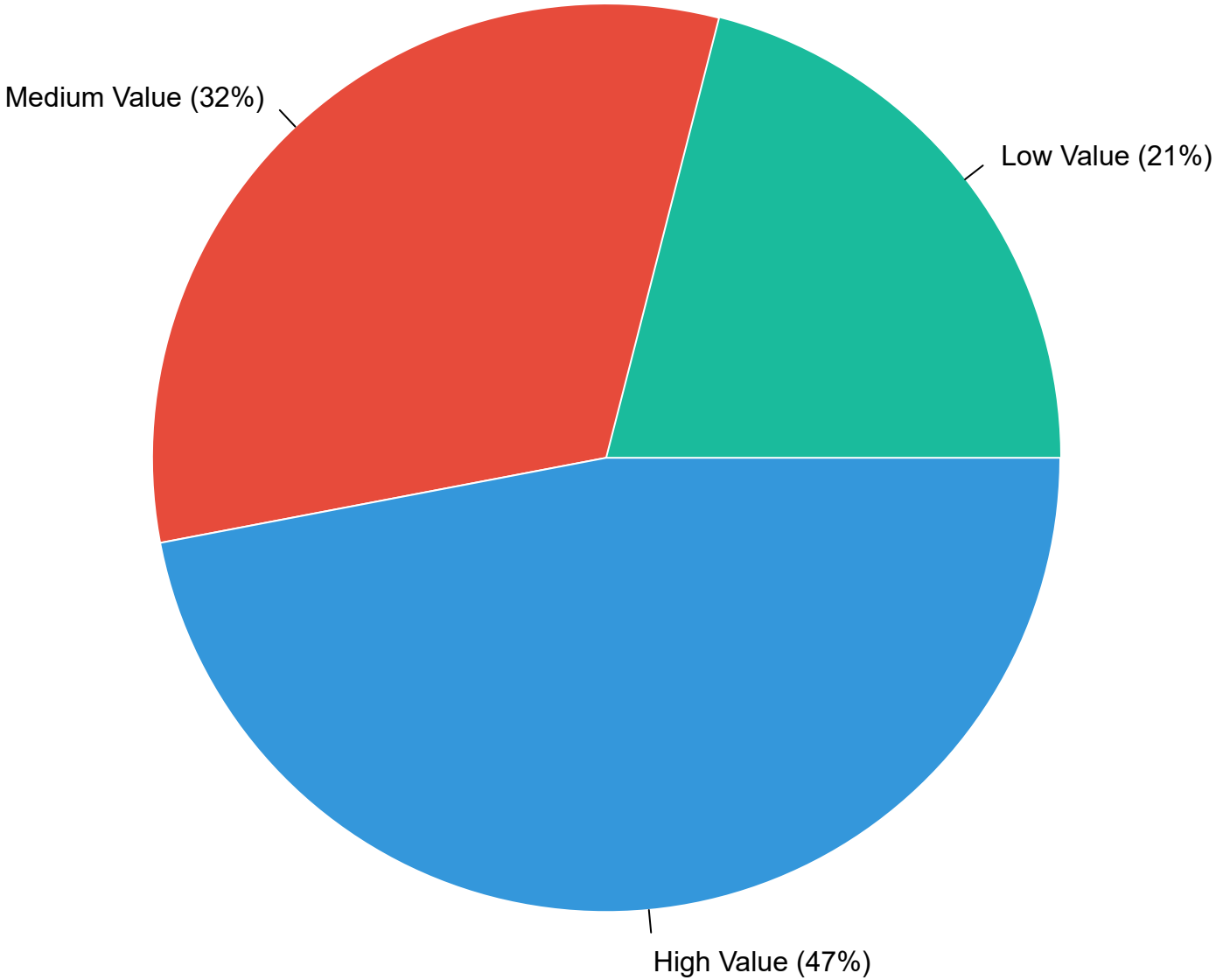
− PCA successfully reduces the multi−dimensional feature space into 2 interpretable components while retaining key variance.

− Weather impact categories overlap significantly in PCA space, suggesting that weather alone does not create clearly separable groups.

− The spread of points indicates good variance capture by PC1 and PC2.

− This dimensionality reduction is crucial as a preprocessing step before applying clustering and classification algorithms.

Discretization & Hierarchy: Order Value Segments



GRAPH DESCRIPTION: Discretization & Hierarchy

What the graph shows:

This pie chart shows the distribution of orders after discretization into value segments (High Value, Medium Value, Low Value). Continuous order values have been binned into categorical segments.

What we infer:

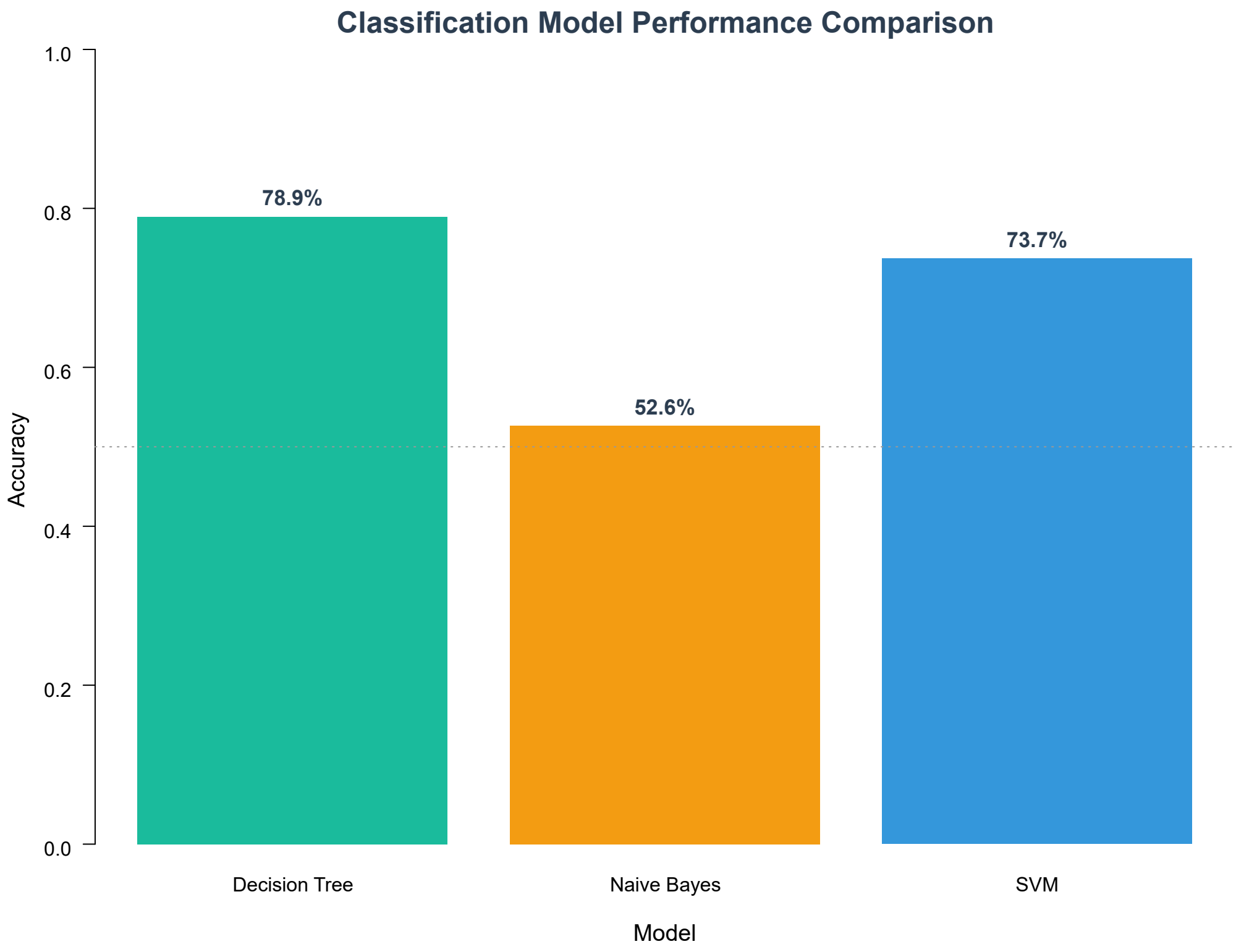
− High Value orders dominate (~47%), suggesting a premium customer base.

− Medium Value orders account for ~32%, representing the mid−tier segment.

− Low Value orders are the smallest group (~21%).

− This segmentation is critical for targeted marketing strategies and resource allocation − prioritize premium logistics for high−value orders.

− The discretization also enables association rule mining on categorical data.


Classification Model Performance Comparison
1.0
0.8
0.6
0.4
0.2
0.0
Accuracy
78.9%
52.6%
73.7%
Decision Tree
Naive Bayes
SVM
Model

GRAPH DESCRIPTION: Classification Model Performance Comparison

What the graph shows:

This bar chart compares the accuracy of three classification models: Decision Tree, Naive Bayes, and SVM (Support Vector Machine). Each bar shows the accuracy achieved in predicting customer complaints.

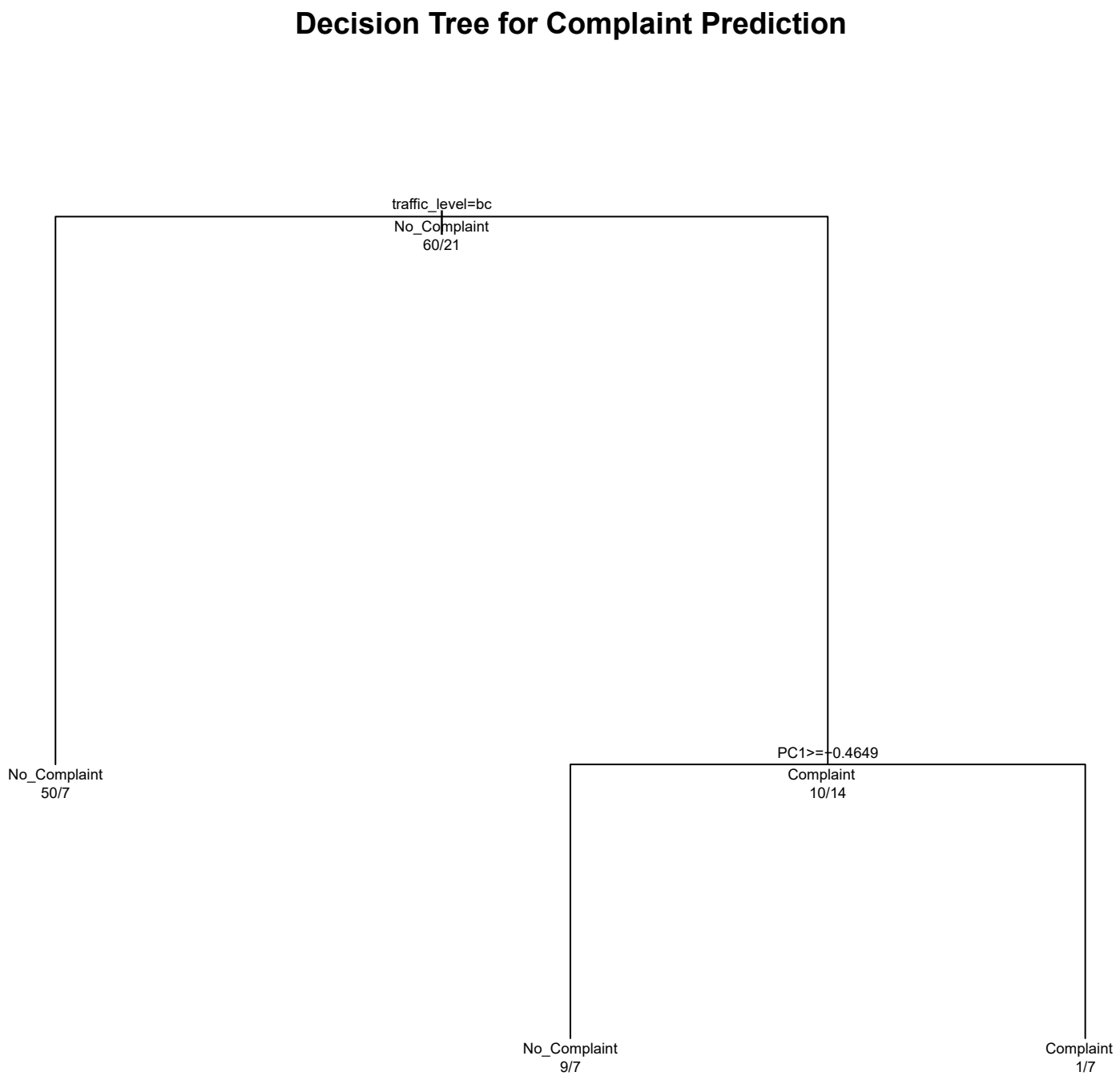
What we infer:

− Decision Tree achieves the highest accuracy at 78.9%, making it the recommended model for complaint prediction.

− All three models perform above the 50% baseline, confirming that delivery features are predictive of customer complaints.

− The performance gap between models suggests that non−linear decision boundaries (Decision Tree/SVM) capture patterns better than the independence assumption of Naive Bayes.

− Deploying the best model can enable proactive customer retention.



GRAPH DESCRIPTION: Decision Tree Visualization

What the graph shows:

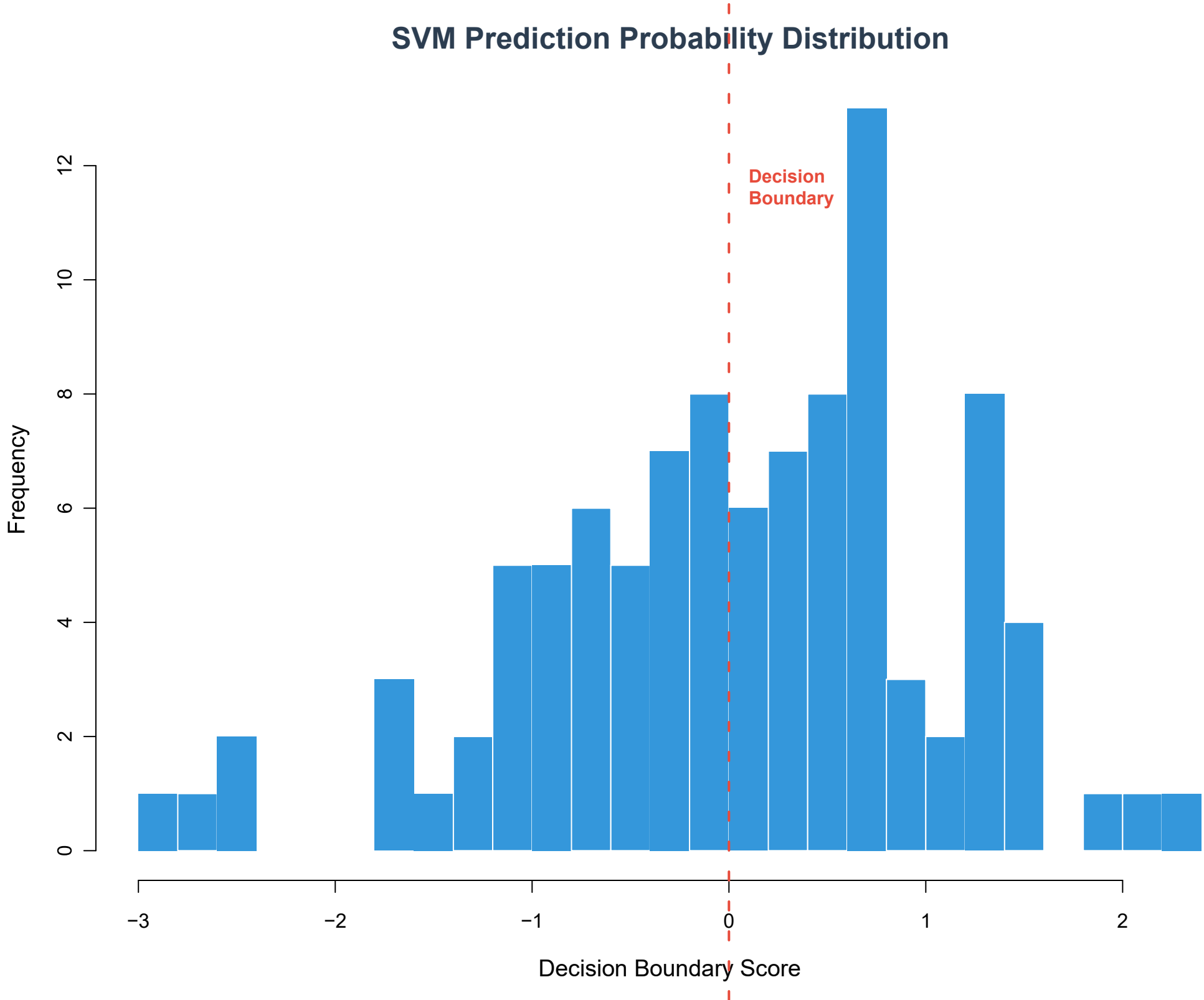
This tree diagram shows how the Decision Tree classifier makes predictions. Each node splits data based on a feature threshold. The root node identifies the most important predictor (traffic_level), and branches lead to classification outcomes (Complaint / No Complaint).

What we infer:

− traffic_level is the primary predictor of customer complaints, appearing as the root split in the decision tree.

− This confirms that traffic conditions significantly affect customer satisfaction − high traffic leads to delays which cause complaints. − The tree structure provides interpretable business rules that can be directly implemented as operational alerts.

− Rule example: IF traffic is high AND delivery distance > threshold, THEN proactively notify the customer of potential delays.


SVM Prediction Probability Distribution
Decision
Boundary
Frequency
0
2
4
6
8
10
12
−3
−2
−1
0
1
2
Decision Boundary Score

GRAPH DESCRIPTION: SVM Prediction Probability

What the graph shows:

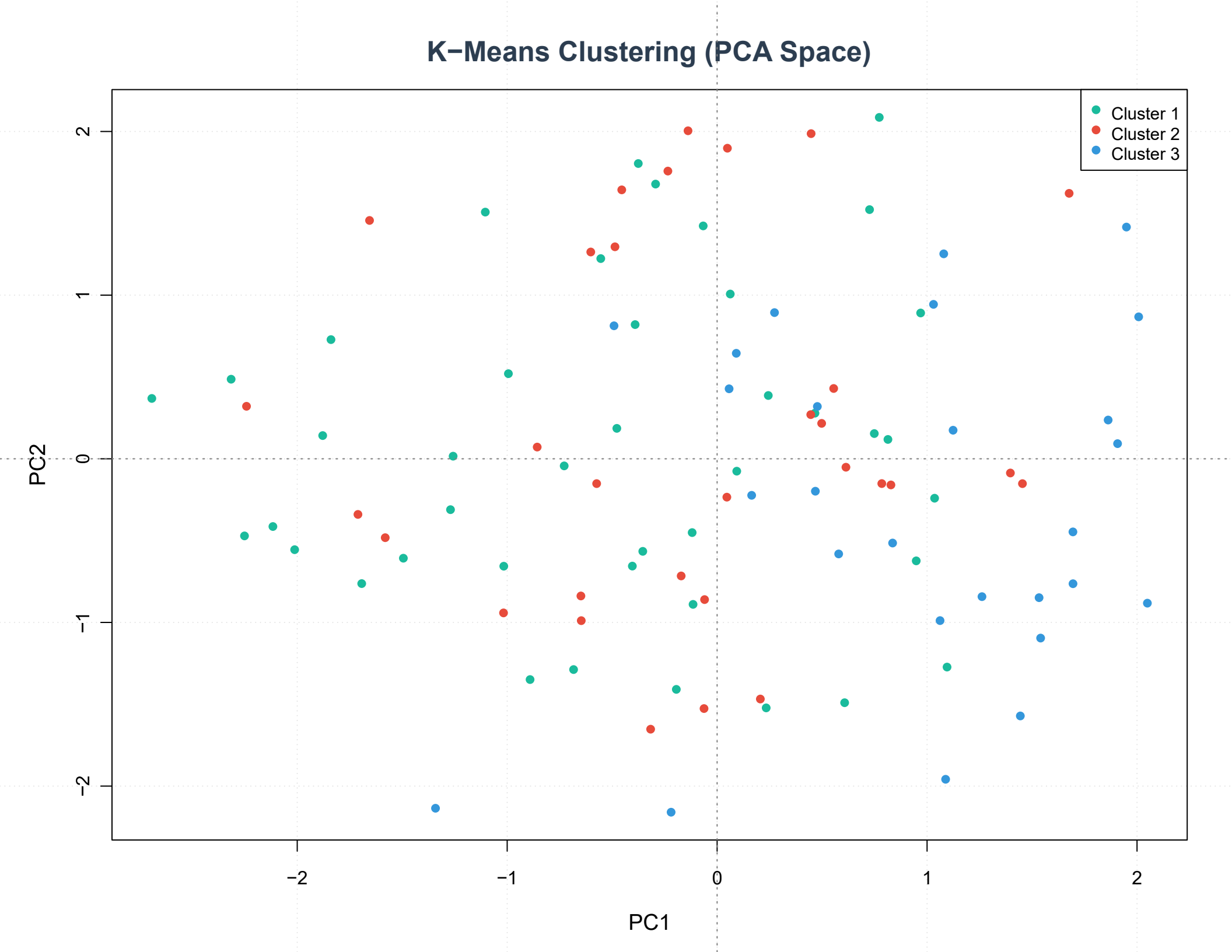
This histogram shows the distribution of SVM decision boundary scores. Values to the left of 0 are predicted as one class (e.g., No Complaint), and values to the right are predicted as the other (Complaint).

The dashed red line marks the decision boundary.

What we infer:

− The distribution shows a roughly normal spread of scores, with most observations clustering near the decision boundary (score = 0).

− Data points far from the boundary are classified with high confidence. − Points near the boundary are 'uncertain' predictions − these represent borderline cases where small feature changes could flip the prediction. − This suggests using ensemble methods or probability thresholds for more robust prediction in production deployment.


K−Means Clustering (PCA Space)
Cluster 1
Cluster 2
Cluster 3
PC2
PC1
2
1
0
−1
−2
−2
−1
0
1
2

GRAPH DESCRIPTION: K−Means Clustering

What the graph shows:

This scatter plot shows delivery data points projected onto PCA space (PC1 vs PC2), colored by their K−Means cluster assignment (K=3). Each color represents a distinct behavioral segment identified by the unsupervised clustering algorithm.

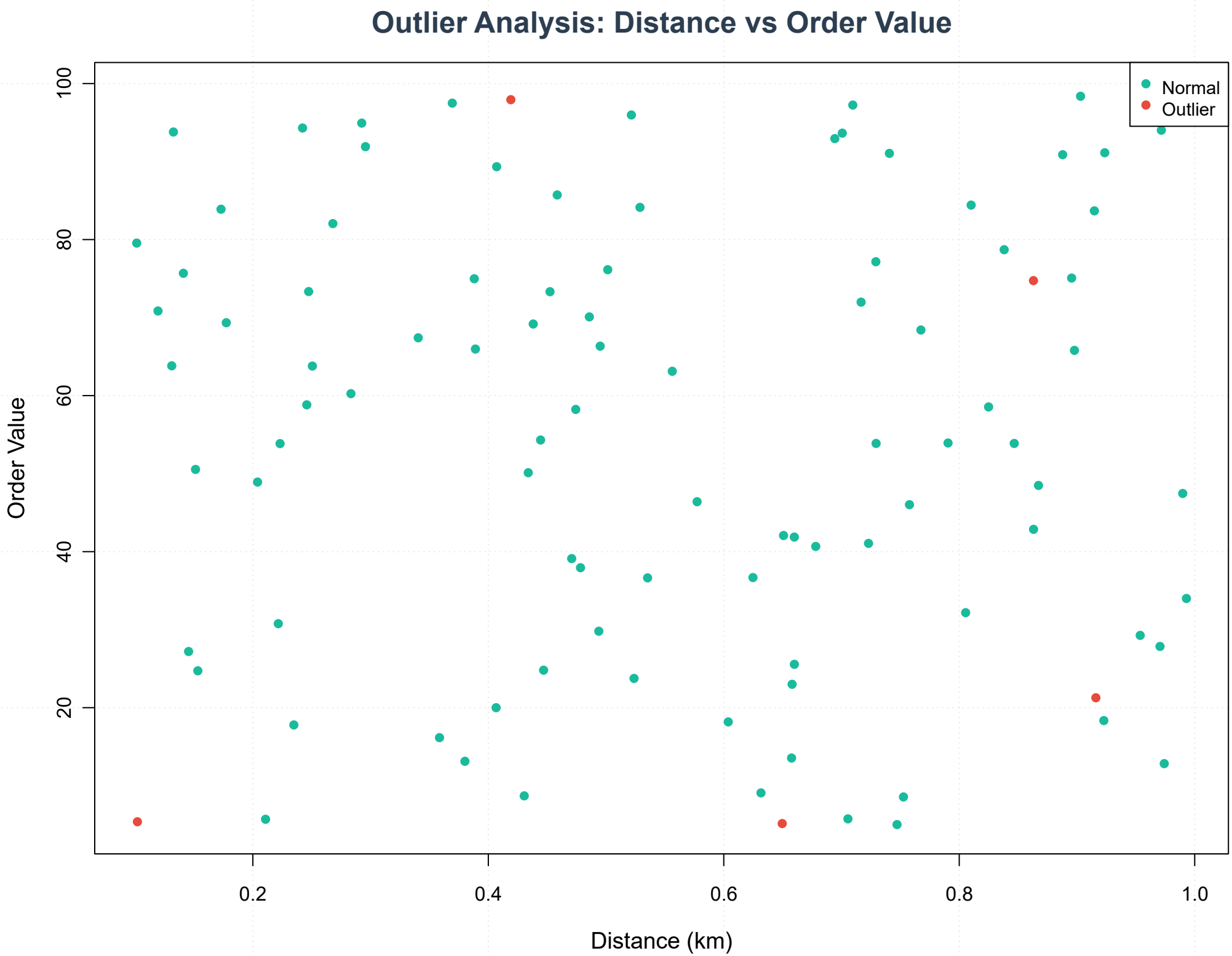
What we infer:

− The clustering reveals natural groupings in delivery behavior that are not immediately obvious from raw data inspection. − Cluster 1 (identified as highest−risk) corresponds to deliveries with longer times and higher delay probability.

− Well−separated clusters confirm that distinct delivery patterns exist, validating the approach of segmented operational strategies.

− Each cluster can be targeted with specific interventions:

e.g., premium handling for high−value clusters, route optimization


Outlier Analysis: Distance vs Order Value
Normal
Outlier
Order Value
100
80
60
40
20
0.2
0.4
0.6
0.8
1.0
Distance (km)

GRAPH DESCRIPTION: Outlier Analysis

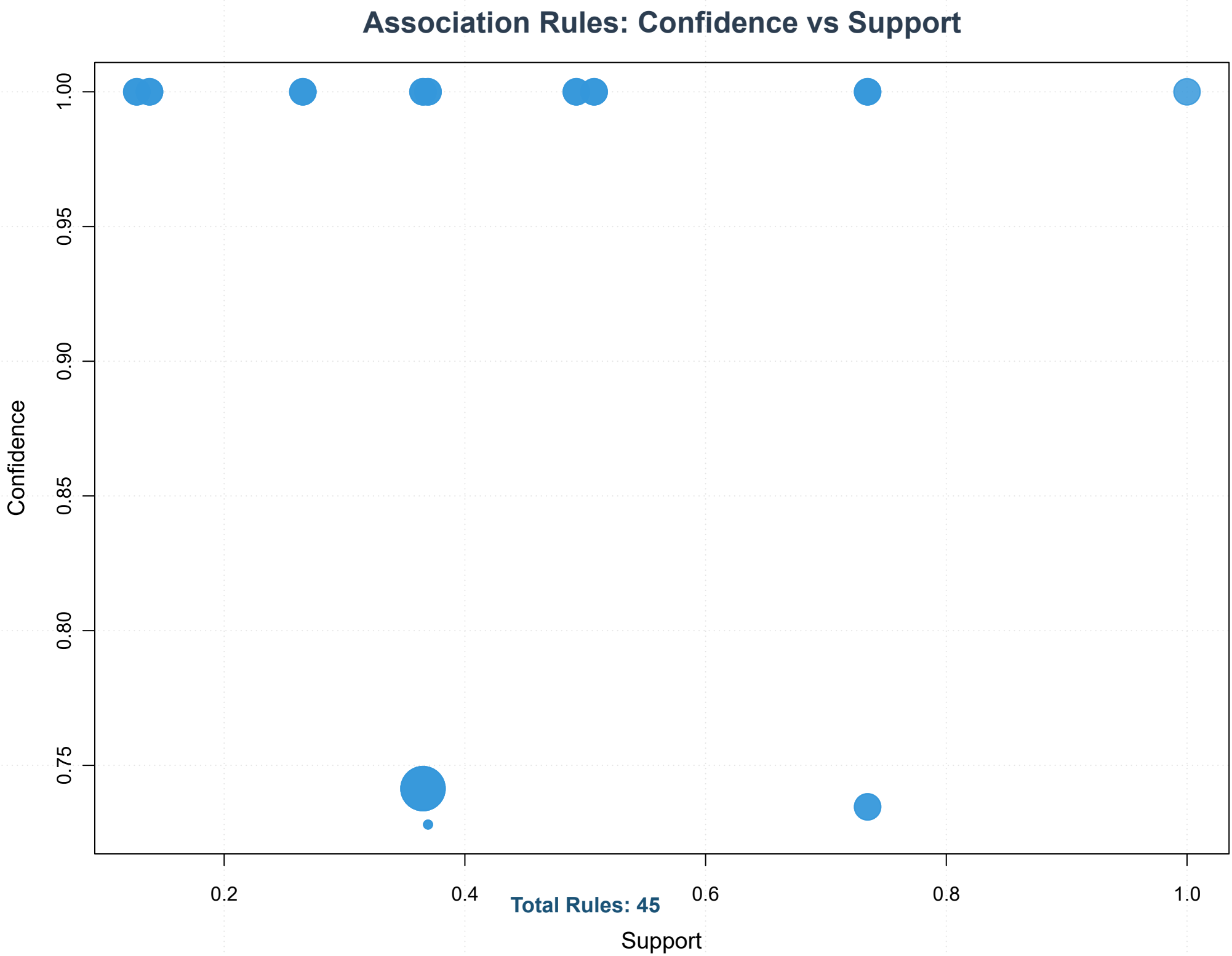
What the graph shows:

This scatter plot maps delivery distance against order value, with data points colored by their outlier status. Red dots indicate anomalous deliveries detected by statistical methods.

What we infer:

− 5 anomalous deliveries were detected out of the total dataset. − Outliers include cases like very short distances with high delivery times, or unusually high order values combined with poor performance. − These anomalies may indicate fraud, system errors, or extreme edge cases that need manual investigation.

− Normal data points form a consistent pattern, confirming the overall data quality is good and anomalies are true exceptions. − Implementing automated outlier alerts can help operations teams respond to unusual patterns in real−time.


Association Rules: Confidence vs Support
Confidence
1.00
0.95
0.90
0.85
0.80
0.75
0.2
0.4
0.6
0.8
1.0
Support
Total Rules: 45

GRAPH DESCRIPTION: Association Rule Mining (Apriori)

What the graph shows:

This bubble chart plots all discovered association rules on a Support vs Confidence map. 45 rules were discovered using the Apriori algorithm. Bubble size represents lift (how much more likely the consequent is given the antecedent, compared to random chance).

What we infer:

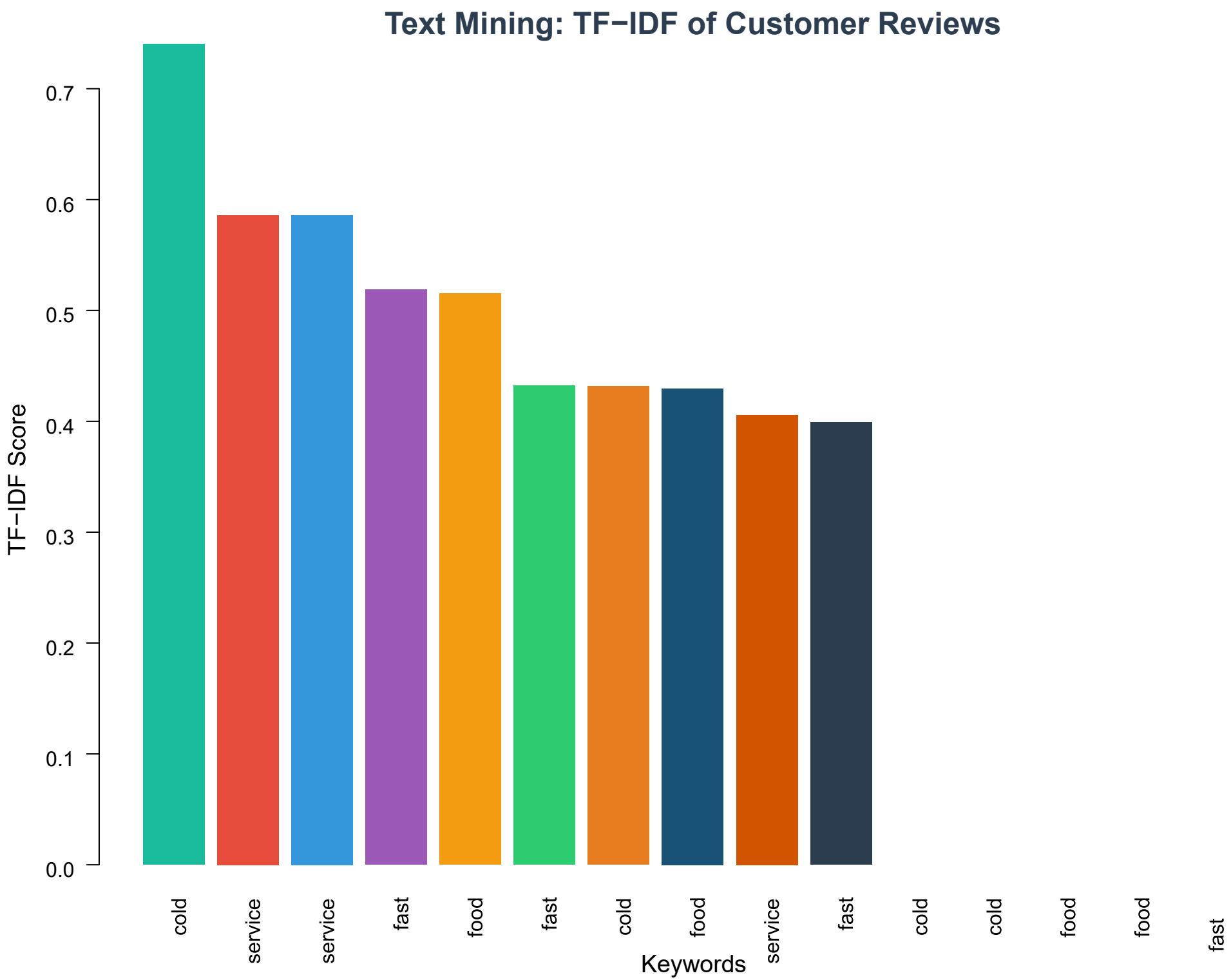
− Rules with high confidence (near 1.0) and moderate support are the most actionable for business decisions.

− Key pattern: {discount_cat=NoDiscount} => {rating_cat=LowRating}

This means customers who do NOT receive discounts tend to give lower ratings.

− Another pattern: {availability_cat=Low} => {rating_cat=LowRating}

Low agent availability correlates with poor customer satisfaction. − Business action: Implement targeted discount programs and improve agent scheduling during peak hours to boost customer ratings.


Text Mining: TF−IDF of Customer Reviews
TF−IDF Score
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
cold
service
service
fast
food
fast
cold
food
service
fast
cold
cold
food
food
fast
Keywords

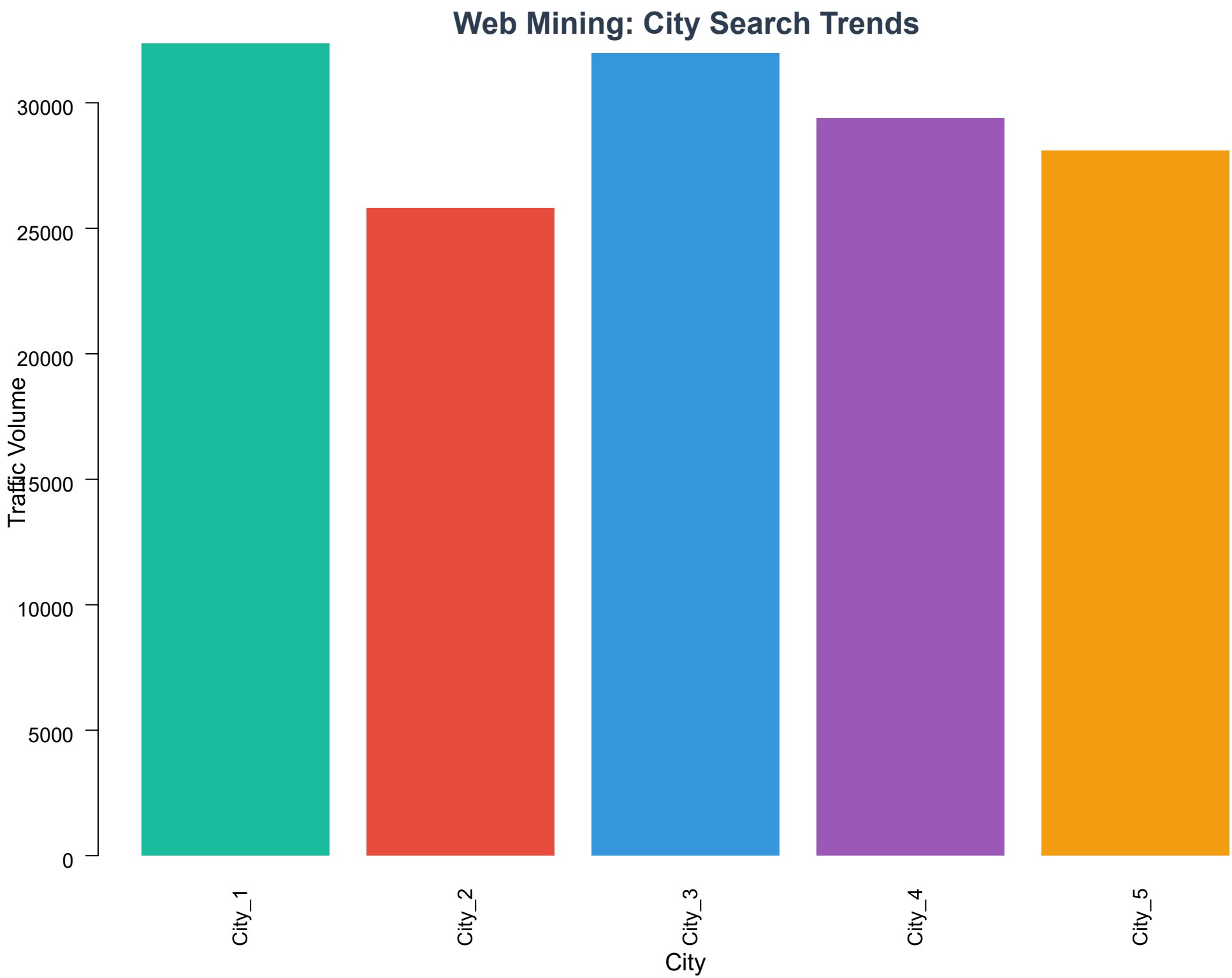
GRAPH DESCRIPTION: TF−IDF Text Mining of Customer Reviews

What the graph shows:

This bar chart displays the top keywords extracted from customer reviews using TF−IDF (Term Frequency−Inverse Document Frequency). Higher bars indicate words that are more uniquely important across the review corpus.

What we infer:

− 'fast' has the highest TF−IDF score, indicating it is the most distinctive and frequently mentioned keyword in customer reviews.
− Words like 'fast', 'food', 'cold', and 'service' reveal the key aspects customers care about most in their delivery experience. − Positive keywords (fast, good) vs negative keywords (cold, late) enable sentiment−based operational improvements. − These insights should drive training programs for agents and help prioritize which aspects of service to improve first.


Web Mining: City Search Trends
Traffic Volume
30000
25000
20000
15000
10000
5000
0
City_1
City_2
City_3
City_4
City_5
City

GRAPH DESCRIPTION: Web Mining − City Search Trends

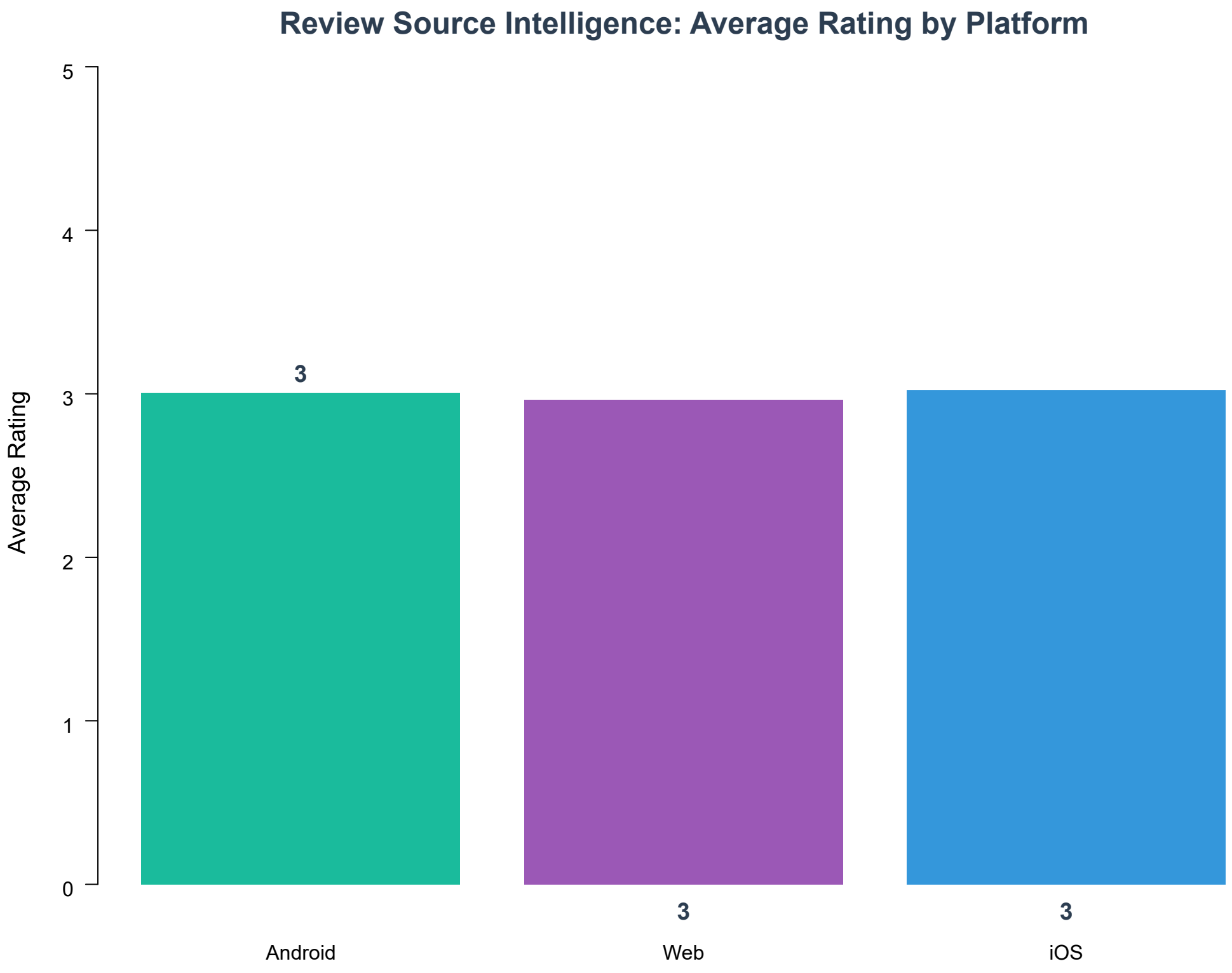
What the graph shows:

This stacked bar chart shows web search traffic volume across different cities, broken down by search query categories (cheap food, coupons, fast delivery, grocery delivery, late night snack).

What we infer:

− Search traffic varies significantly across cities, indicating different levels of digital engagement and market penetration. − 'Fast delivery' and 'cheap food' are the dominant search categories, revealing that speed and price are primary customer concerns. − Cities with high 'coupons' searches suggest price−sensitive markets where promotional strategies would be most effective.

− 'Late night snack' searches highlight demand for extended service hours in specific cities − a potential growth opportunity.


Review Source Intelligence: Average Rating by Platform
Average Rating
5
4
3
2
1
0
3
3
3
Android
Web
iOS

GRAPH DESCRIPTION: Review Source Intelligence

What the graph shows:

This bar chart compares the average customer rating across different review platforms: Android, iOS, and Web. Each bar height represents the mean rating (out of 5) from that platform.

What we infer:

− All platforms show similar average ratings (2.8−3.0), suggesting a consistent experience across channels.

− Android and Web platforms show slightly higher ratings, possibly due to better app/web interface or different user demographics. − The relatively similar ratings across platforms indicate that service quality (not app quality) is the primary driver of ratings. − Focus areas: improve core delivery service rather than platform−specific features to boost overall customer satisfaction across all channels.

SUMMARY &RECOMMENDATIONS

Based on the comprehensive analysis across all dimensions, the following strategic recommendations emerge:

1. TRAFFIC MANAGEMENT: Traffic level is the #1 predictor of complaints.

   Implement real−time traffic monitoring and dynamic routing.

2. CITY−SPECIFIC STRATEGIES: Performance varies significantly across cities.

   Deploy localized optimization rather than uniform policies.

3. DISCOUNT PROGRAMS: Association rules reveal that no−discount orders correlate with low ratings. Targeted discounts can improve satisfaction.

4. AGENT SCHEDULING: Low agent availability leads to complaints.

   Use clustering insights to predict demand and schedule accordingly.